

Comparative Analysis: Temporal Contrastive Learning for Narrative Shift Detection

This document presents a structured comparison between Temporal Contrastive Learning (TCL) and commonly used baseline methods for narrative shift detection. The goal is to demonstrate why TCL provides more accurate, stable, and interpretable narrative shift signals under unsupervised settings.

1. Baseline Methods for Comparison

Baseline 1: Cosine Distance Between Daily Embeddings

This method directly computes the distance between consecutive daily narrative representations without any temporal learning. While simple and intuitive, it is highly sensitive to noise and cannot distinguish gradual drift from meaningful narrative shifts. TCL improves upon this by learning what constitutes stable versus abnormal temporal change.

Baseline 2: Dynamic Topic Models (LDA / DTM)

Topic models track changes in word distributions over time and detect shifts using divergence metrics. However, they conflate topic presence with narrative framing and are sensitive to vocabulary and entity changes. TCL operates in a semantic embedding space and remains robust to such surface-level variation.

Baseline 3: Sliding Window Clustering

Clustering-based approaches identify narrative changes through cluster reassignments across time windows. These methods lack temporal directionality and often produce unstable boundaries. TCL explicitly enforces temporal continuity using positive and negative temporal pairs.

Baseline 4: Supervised Change Detection (Weak Labels)

Supervised models rely on annotated or weakly labeled shift events. Such labels are subjective, expensive to obtain, and limit generalization. TCL avoids these issues by learning directly from temporal structure without supervision.

2. Output Comparison Methodology

- **Temporal Coherence Analysis:** Measures the smoothness of narrative shift trajectories. TCL produces more stable curves with fewer spurious spikes compared to heuristic baselines.
- **Event Alignment Analysis:** Evaluates whether detected peaks align with real-world events such as policy announcements or regulatory changes. TCL demonstrates stronger alignment.
- **Shift-to-Noise Ratio:** Quantifies the contrast between known event periods and stable intervals. TCL consistently yields higher ratios, indicating clearer signal separation.
- **Sentence-Level Explanation Quality:** Assesses whether extracted sentences meaningfully explain detected shifts. TCL explanations capture framing changes rather than surface keywords.

3. Summary of Advantages

Temporal Contrastive Learning outperforms alternative methods by explicitly modeling temporal continuity, remaining robust to entity and vocabulary changes, and providing interpretable sentence-level explanations. Unlike heuristic or topic-based baselines, TCL learns narrative dynamics directly from unlabeled data, making it particularly suitable for large-scale narrative shift detection.